



U.S. General Services
Administration

Chapter 3.10: Efficiency Metrics

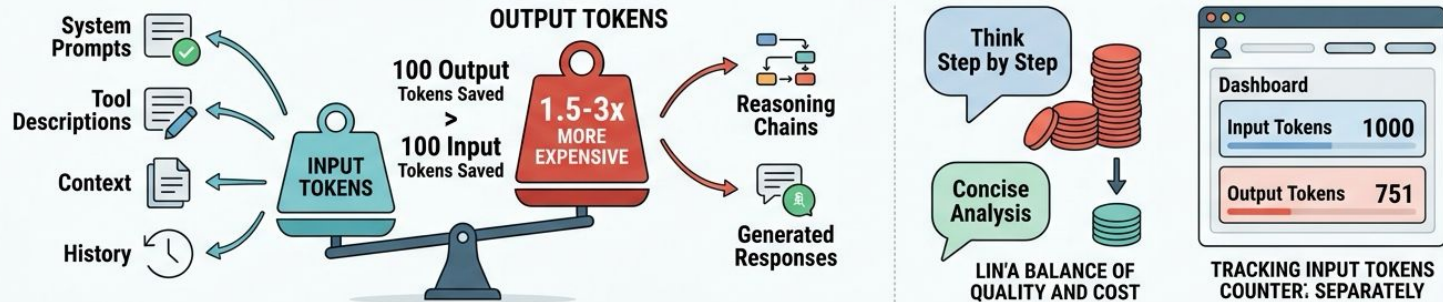
US General Service Administration
AI Community of Practice - March 2026

Token Economics

MAXIMIZING AGENT SURVIVAL THROUGH EFFICIENCY



UNDERSTANDING TOKEN ECONOMICS: INPUT VS. OUTPUT ASYMMETRY



The Hidden Token Sinks

- Tool descriptions consume thousands of tokens per request
 - Conversation history bloats over multi-turn interactions
 - Retrieved context often pulls far more than needed
 - AgentDiet framework: 40-60% input reduction, same accuracy
 - Three waste types: useless, redundant, expired information
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Workflow Architecture -- Where Costs Multiply or Collapse

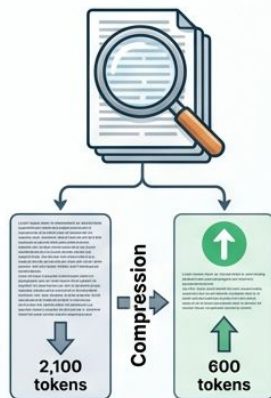
- Sequential agent handoffs duplicate context at each step
 - Parallel orchestration reduces tokens 40-60%
 - Smart routing: small models handle 80% of queries cheaply
 - Escalate to frontier models only for complex reasoning
 - Graceful degradation prevents runaway token spending
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Core Metrics -- The Measurement Framework

- Total Token Consumption: input + output tracked separately
 - Token Efficiency Ratio: quality per token consumed
 - Cost Per Interaction: direct financial attribution
 - Step Reduction: fewer steps without accuracy loss
 - API Call Frequency: calls per task, target redundancy
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- A decorative graphic in the bottom right corner consisting of several overlapping, semi-transparent blue rectangular blocks of varying sizes and orientations, creating a modern, abstract look.

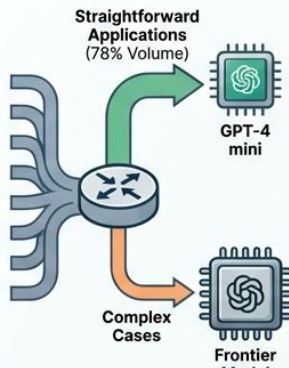
Optimization Strategies

Compress Tool Descriptions



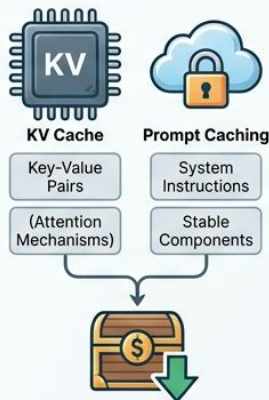
Same information, significantly fewer tokens.

Model Routing



➔ **62% token reduction** in loan processing

KV & Prompt Caching



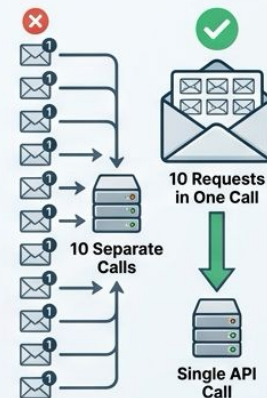
Cut costs 15-30% by skipping repeated computations.

Trajectory Pruning



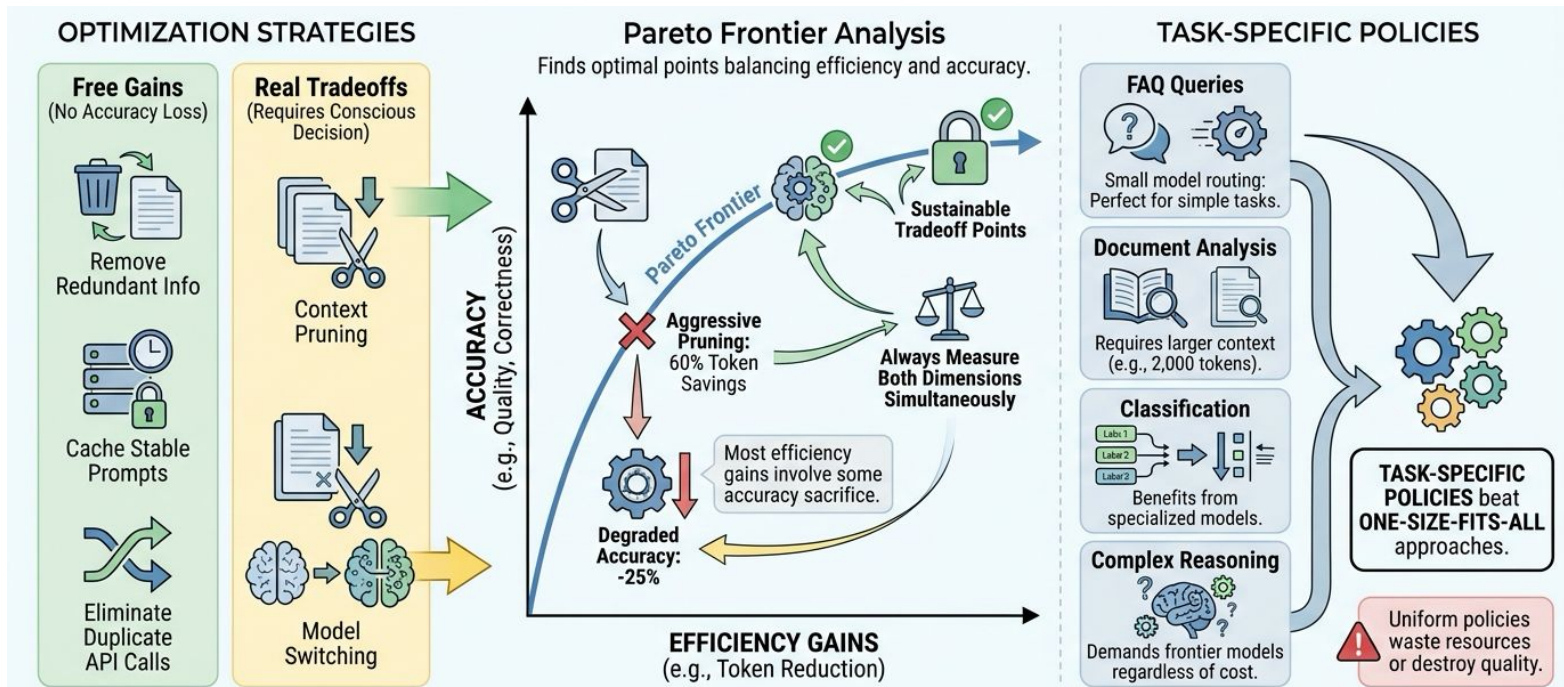
Removes dead-end reasoning waste, redundant restatements, and expired context.

Batch Operations

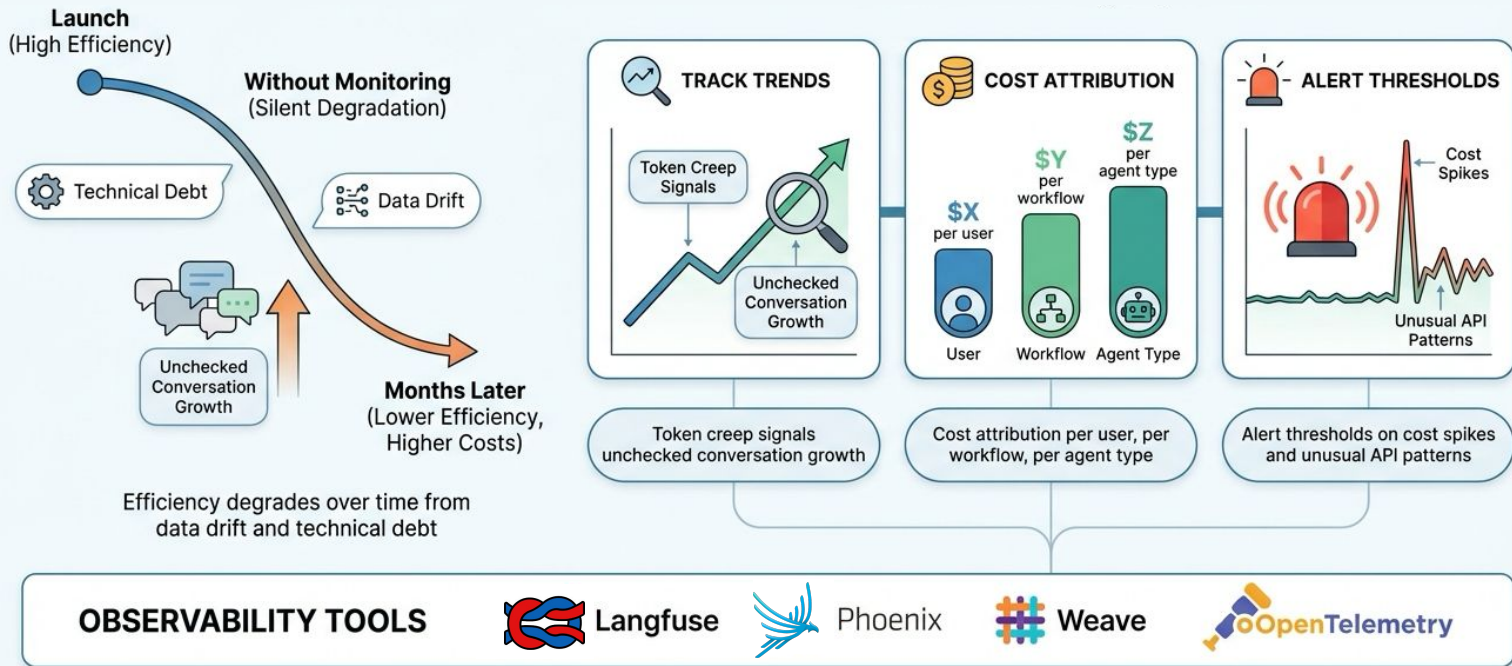


Dramatically better throughput and lower per-operation overhead.

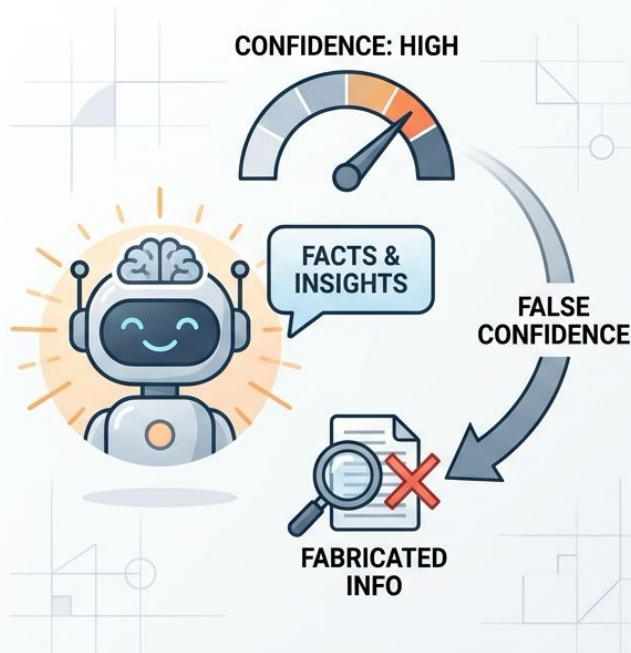
The Efficiency-Accuracy Tradeoff



Production Monitoring



Hallucination Rate

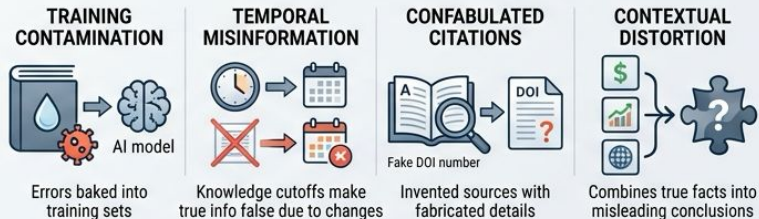


References: Neural Uncertainty Principle: Hallucination Detection (2026)
EvidenceRL: Reinforcing Evidence Consistency (2026)

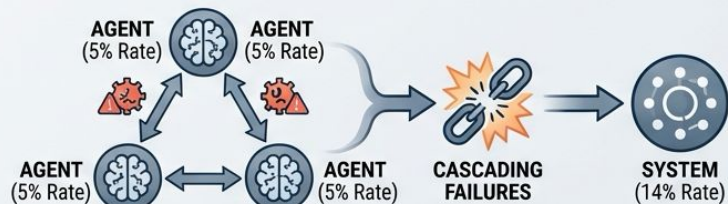
PRODUCTION HALLUCINATION RATES



FOUR TYPES OF HALLUCINATIONS



MULTI-AGENT SYSTEMS COMPOUND RATES

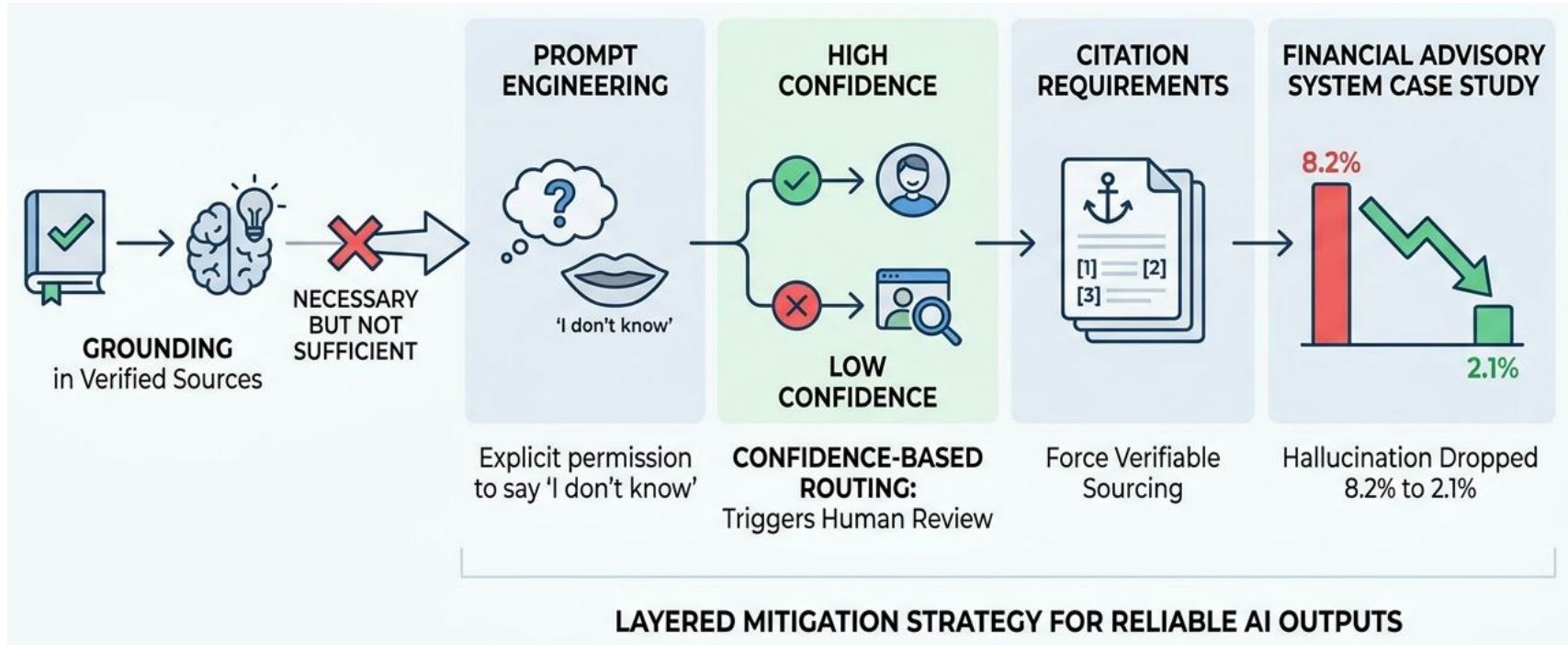


Individual 5% rates produce 14% system-level rates through propagation

Multi-Layer Hallucination Detection

- Stage 1: Deterministic validators catch structural fabrications
 - Stage 2: Statistical evaluators flag low-confidence outputs
 - Stage 3: Semantic validators verify grounding via entailment
 - Stage 4: LLM-as-judge for nuanced assessment of flagged cases
 - Multi-method catches 40-60% more than any single technique
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Mitigation That Actually Works



Key Takeaways

- Architecture matters more than micro-optimizations
 - Always track efficiency and accuracy together
 - Hallucination mitigation requires defense in depth, not a silver bullet
 - Continuous monitoring prevents silent cost and quality degradation
 - Next up: Part 4 takes your optimized agents into production deployment
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